

CLASS 9 · CBSE · GANITA MANJARI PART 2 · SHORT NOTES

# Chapter 12: Quadrilaterals

Topic-wise short notes with solved examples

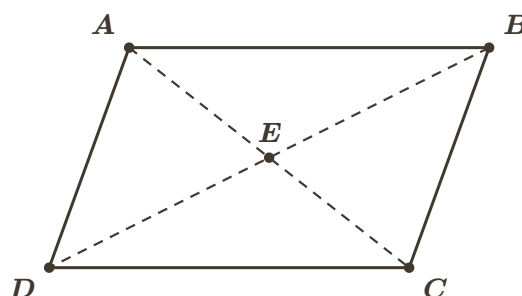
This chapter defines a quadrilateral carefully, proves the tests for a parallelogram, and uses parallelograms to prove the Midpoint Theorem, the Centroid Theorem and Varignon's theorem. It ends with tiling the plane using any quadrilateral.

## 1 What a quadrilateral is

- A **quadrilateral** (4-gon)  $ABCD$  is formed by four points  $A, B, C, D$  in a plane, no three of them collinear, joined by the segments  $AB, BC, CD$  and  $DA$ , where every point other than the four vertices lies on exactly one of these segments.
- This rules out three collinear vertices (which would give a triangle), four points not in one plane (a non-planar figure) and sides that cross each other (a self-intersecting figure).
- **Sides:**  $AB, BC, CD, DA$ . **Diagonals:**  $AC$  and  $BD$ . **Adjacent** sides share a vertex and **opposite** sides do not. Angles at the ends of a side are adjacent angles.
- The same quadrilateral can be named  $BCDA, CDAB, DABC$  or in the reverse order, but not  $ACBD$ .
- A quadrilateral is **convex** when all its interior angles are less than  $180^\circ$ . Then its diagonals intersect. A **non-convex** quadrilateral has one angle greater than  $180^\circ$  (a dent), and its diagonals do not intersect.
- The interior angles of any quadrilateral add up to  $360^\circ$ .

## 2 Parallelograms and their properties

- A **parallelogram** is a quadrilateral whose opposite sides are parallel. Rectangles, rhombuses and squares are parallelograms.
- **Theorem 1:** In a parallelogram (a) opposite sides are equal (b) opposite angles are equal (c) the diagonals bisect each other.



$ABCD$  is a parallelogram.  $E$  is the mid-point of both diagonals.

- (a) The diagonal  $AC$  is a transversal for both pairs of parallel sides, so  $\triangle ACD \cong \triangle CAB$  by ASA. Hence  $AB = DC$  and  $AD = BC$ .
- (b) Adjacent angles are interior angles on the same side of a transversal, so they add up to  $180^\circ$ . It follows that  $\angle A = \angle C$  and  $\angle B = \angle D$ .
- (c)  $\triangle AED \cong \triangle CEB$  (using  $AD = BC$  and the alternate angles), so  $EA = EC$  and  $EB = ED$ .

**Example 2.1** In a parallelogram  $ABCD$ ,  $\angle A = (3x + 10)^\circ$  and  $\angle B = (2x + 20)^\circ$ . Find all four angles.

SOLUTION

Adjacent angles are supplementary:  $(3x + 10) + (2x + 20) = 180$ , so  $x = 30$ .

$\angle A = \angle C = 100^\circ$  and  $\angle B = \angle D = 80^\circ$

**Example 2.2** The diagonals of a parallelogram  $ABCD$  meet at  $E$ . If  $AC = 16$  cm and  $BD = 10$  cm, find  $AE$  and  $BE$ .

SOLUTION

The diagonals bisect each other.

$AE = 8$  cm and  $BE = 5$  cm

### 3 Tests for a parallelogram

- The converses of the three properties are all true, and there is one more useful test.

| Test      | If a quadrilateral has ... then it is a parallelogram |
|-----------|-------------------------------------------------------|
| Theorem 2 | both pairs of opposite sides equal                    |
| Theorem 3 | both pairs of opposite angles equal                   |
| Theorem 4 | diagonals that bisect each other                      |
| Theorem 5 | one pair of opposite sides both equal and parallel    |

- Proof of Theorem 2:  $\triangle ACD \cong \triangle CAB$  by SSS, so the alternate angles are equal and the opposite sides are parallel.
- Proof of Theorem 3:  $2(\angle A + \angle B) = 360^\circ$ , so  $\angle A + \angle B = 180^\circ$  and  $AD \parallel BC$ . In the same way  $AB \parallel DC$ .
- Proof of Theorem 4:  $\triangle AED \cong \triangle CEB$  by SAS, which gives equal alternate angles.
- Proof of Theorem 5: if  $AB \parallel DC$  and  $AB = DC$ , then  $\triangle EAB \cong \triangle ECD$  by ASA, so the diagonals bisect each other. Now use Theorem 4.
- Special parallelograms: a parallelogram with one right angle is a **rectangle**. A parallelogram with equal diagonals is a rectangle. A parallelogram whose diagonal bisects an angle is a **rhombus**.

**Example 3.1** Show that  $A(1, 1)$ ,  $B(5, 2)$ ,  $C(7, 6)$  and  $D(3, 5)$  are the vertices of a parallelogram.

SOLUTION

Mid-point of  $AC = \left(\frac{1+7}{2}, \frac{1+6}{2}\right) = (4, 3.5)$ . Mid-point of  $BD = \left(\frac{5+3}{2}, \frac{2+5}{2}\right) = (4, 3.5)$ .

**The diagonals bisect each other, so  $ABCD$  is a parallelogram (Theorem 4).**

**Example 3.2** One pair of opposite sides of a quadrilateral are equal and the other pair are parallel. Must it be a parallelogram?

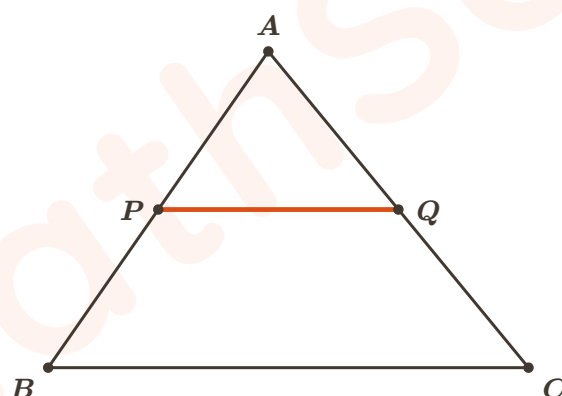
SOLUTION

Theorem 5 needs the **same** pair of sides to be equal and parallel.

**No. An isosceles trapezium is a counterexample.**

## 4 The Midpoint Theorem

- Theorem 6 (Midpoint Theorem):** The segment joining the mid-points of two sides of a triangle is parallel to the third side and is half as long.



$P$  and  $Q$  are mid-points.  $PQ$  is parallel to  $BC$  and  $PQ = \frac{1}{2} BC$ .

- Proof idea: draw a line through  $C$  parallel to  $BA$ , meeting  $PQ$  produced at  $R$ . Then  $\triangle APQ \cong \triangle CRQ$  (AAS), so  $PQ = QR$  and  $CR = AP = BP$ . Since  $CR$  is equal and parallel to  $BP$ ,  $BCRP$  is a parallelogram (Theorem 5). Hence  $PQ \parallel BC$  and  $PQ = \frac{1}{2} PR = \frac{1}{2} BC$ .
- Theorem 7 (Converse):** The line through the mid-point of one side of a triangle, parallel to another side, bisects the third side.
- Joining the mid-points of all three sides divides a triangle into **four congruent triangles**. Each has sides half as long as the original.

**Example 4.1** The sides of a triangle are 10 cm, 14 cm and 18 cm. Find the perimeter of the triangle formed by joining the mid-points of its sides.

SOLUTION

Each side of the new triangle is half of one side: 5 cm, 7 cm and 9 cm.

**21 cm**

**Example 4.2** In  $\triangle ABC$ ,  $P$  is the mid-point of  $AB$  and the line through  $P$  parallel to  $BC$  meets  $AC$  at  $Q$ . If  $AC = 13$  cm and  $BC = 12$  cm, find  $AQ$  and  $PQ$ .

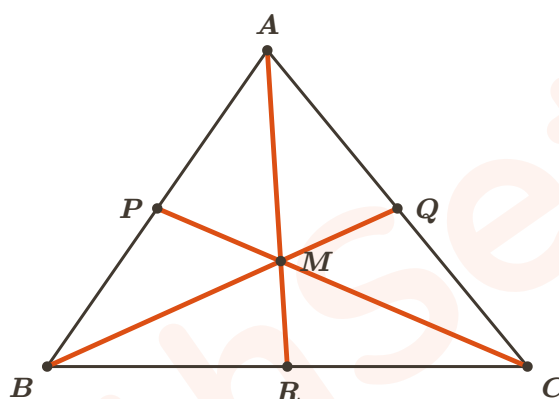
SOLUTION

By the converse of the Midpoint Theorem,  $Q$  is the mid-point of  $AC$ .

$AQ = 6.5$  cm and  $PQ = 6$  cm

## 5 Medians and the centroid

- A **median** joins a vertex of a triangle to the mid-point of the opposite side.
- Theorem 8 (Centroid Theorem):** The three medians of a triangle pass through one point, the **centroid**. It divides each median in the ratio  $2 : 1$ , with the longer part towards the vertex.



$M$  is the centroid.  $AM : MR = 2 : 1$ .

- Proof idea: let two medians  $CP$  and  $BQ$  meet at  $M$ , and let  $X, Y$  be the mid-points of  $BM$  and  $CM$ . By the Midpoint Theorem in  $\triangle ABC$  and in  $\triangle MBC$ , both  $PQ$  and  $XY$  are parallel to  $BC$  and equal to half of it. So  $\triangle MPQ \cong \triangle MYX$ , which gives  $MP = MY = YC$  and  $MQ = MX = XB$ . The same argument with another pair of medians gives the same point.

**Example 5.1** A median of a triangle is 12 cm long. How far is the centroid from the vertex and from the mid-point of the opposite side?

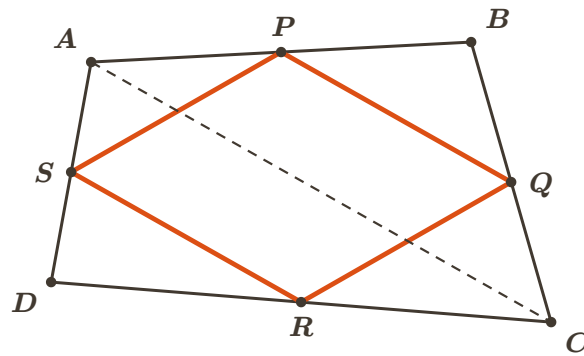
SOLUTION

The centroid divides the median in the ratio  $2 : 1$ :  $\frac{2}{3} \times 12$  and  $\frac{1}{3} \times 12$ .

**8 cm from the vertex and 4 cm from the mid-point**

## 6 The Midpoint Theorem for quadrilaterals

- Theorem 9 (Varignon's theorem):** The mid-points of the four sides of any quadrilateral are the vertices of a parallelogram, called the **Varignon parallelogram**.



$PQ$  and  $SR$  are both parallel to the diagonal  $AC$

- Proof: by the Midpoint Theorem in  $\triangle ABC$  and  $\triangle ADC$ , both  $PQ$  and  $SR$  are parallel to  $AC$  and equal to half of it. So  $PQ$  is equal and parallel to  $SR$ .
- The sides of the Varignon parallelogram are half the diagonals. So its perimeter equals the sum of the two diagonals of the quadrilateral.
- If the diagonals are **equal**, it is a rhombus. If they are **perpendicular**, it is a rectangle.

**Example 6.1** The diagonals of a quadrilateral are 12 cm and 16 cm. Find the sides and the perimeter of its Varignon parallelogram.

SOLUTION

The sides are half the diagonals: 6 cm and 8 cm.

**Sides 6 cm, 8 cm, 6 cm, 8 cm; perimeter 28 cm**

## 7 Tiling the plane

- To **tile** the plane is to cover it with copies of a shape, leaving no gaps and no overlaps.
- Any parallelogram tiles the plane: a grid of two sets of parallel lines does it.
- **Any quadrilateral tiles the plane.** Its four angles add up to  $360^\circ$ , so four copies fit around a point with each angle used once.
- Method: rotate a copy through  $180^\circ$  about the mid-point of a side. It fits exactly along that side. Repeat at every side of every copy.
- In such a tiling the Varignon parallelograms of the copies form a grid of parallelograms.
- A shape can tile around a point only if its angles can add up to  $360^\circ$  there. A regular pentagon cannot tile the plane: each angle is  $108^\circ$ , and 360 is not a multiple of 108.

**Example 7.1** Can a regular hexagon tile the plane?

SOLUTION

Each interior angle of a regular hexagon is  $120^\circ$ , and  $3 \times 120^\circ = 360^\circ$ .

**Yes. Three hexagons meet at each vertex, as in a honeycomb.**

## 8 Common mistakes

- Using Theorem 5 with one pair of sides equal and the **other** pair parallel. The same pair must be both.
- Saying the diagonals of every parallelogram are equal or perpendicular. They only bisect each other.
- Writing the centroid ratio the wrong way round. The part from the vertex is the longer one.
- Assuming the converse of a property without proof. Each converse here has its own proof.
- Naming the vertices out of order:  $ABCD$  means  $A$  is joined to  $B$ ,  $B$  to  $C$ ,  $C$  to  $D$  and  $D$  to  $A$ .

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